

Exp No: 9- ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

Objectives:

1. To interface an ultrasonic sensor with the Arduino UNO.
2. To measure the distance of an object using the ultrasonic sensor.
3. To understand ultrasonic sensor interfacing and distance measurement using Arduino in Proteus.

Program:

// Program Sketch

```
int trig = 10;
```

```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup() {
```

```
// put your setup code here, to run once:
```

```
pinMode(trig, OUTPUT);
```

```
pinMode(echo, INPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() { // Sketch to calculate ultrasonic distance
```

```
digitalWrite(trig, LOW);
```

```
delayMicroseconds(10);
```

```
digitalWrite(trig, HIGH);
```

```

delayMicroseconds(10);

digitalWrite(trig, LOW);

delayMicroseconds(10);

duration = pulseIn(echo, HIGH);

cm = (duration/2) * 0.034;

// Print Value on Serial Monitor

Serial.print(" Distance = ");

Serial.print(cm);

Serial.println(" cm");

}

```

Output:

